

Question Paper

Exam Date & Time: 30-Dec-2025 (09:30 AM - 12:30 PM)



MANIPAL ACADEMY OF HIGHER EDUCATION

MANIPAL INSTITUTE OF TECHNOLOGY - BENGALURU

FIRST SEMESTER M.TECH. DEGREE, MAKEUP EXAMINATIONS - DECEMBER 2025

Analog VLSI Design [ECE_5153]

Marks: 50

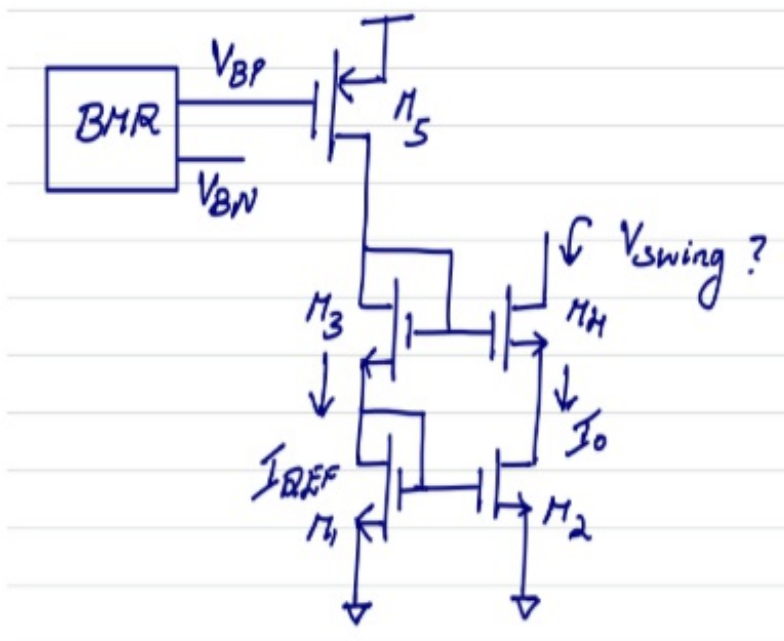
Duration: 180 mins.

PART-A

Answer all the questions.

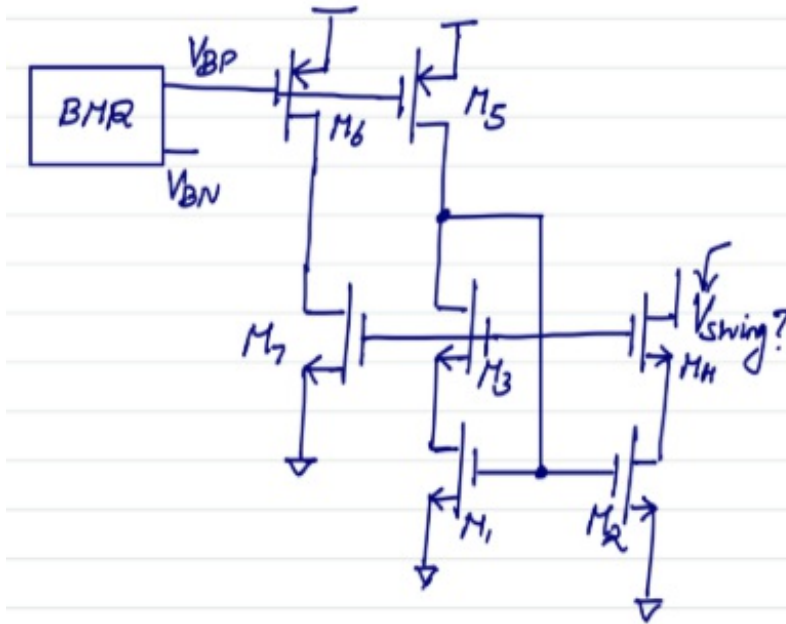
- 1) Discuss the current mirror swings in the below configuration (5)

A)



Course Outcome: "CO1" / Cognitive Level: "Application"

- B) Discuss the current mirror swings in the below configuration (5)



Course Outcome: "CO1" / Cognitive Level: "Application"

- 2) Design a circuit that can self-bias itself (BMR) to generate a constant reference current. Design BMR to give a constant bias current of $30\mu\text{A}$ (4)

A)

Course Outcome: "CO2" / Cognitive Level: "Application"

- B) What are the two stable operating conditions in which the BMR can stably operate and are both the states desirable? Explain (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- C) Design a circuit to nudge the BMR to start in case reference current becomes zero. (3)

Course Outcome: "CO2" / Cognitive Level: "Application"

- 3) Design a differential amplifier circuit to have a gain of at least 20V/V to reduce the sensitivity to V_{DD} for the BMR designed in Question 2. Come up with the total BMR circuit inclusive of startup and differential amplifier to regulate I_{REF} (10)

Course Outcome: "CO3" / Cognitive Level: "Analysis"

- 4) For a given G_m in a common source configuration how to achieve the maximum possible gain by changing R_{out} . Discuss both the theoretical and practical limits to the gain that is achievable. (10)

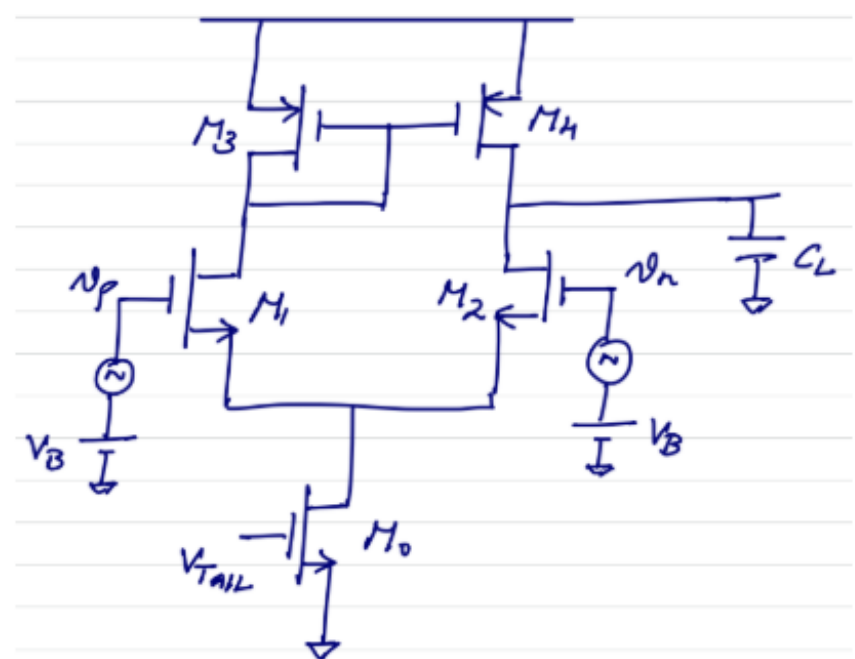
Course Outcome: "CO4" / Cognitive Level: "Analysis"

- 5) (4)

A)

Design a differential input single ended output amplifier with a maximum load capacitance of 1pF to have the following. (For small signal analysis consider the tail transistor M0 to be almost ideal current source).

Gain > 40db, BW > 300kHz. Power < 120μW



Course Outcome: "CO5" / Cognitive Level: "Analysis"

- B) Use the BMR designed in Question no. 2 to bias the MOSFET in saturation. (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

- C) Can you discuss a method to cascade this differential amplifier with a common source amplifier to increase the gain by another 20dB. What will the consequence on the order of the system and the bandwidth? (3)

Course Outcome: "CO5" / Cognitive Level: "Analysis"

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Event Name		Subject	
058-MIT BLR-MAKEUP EXAM PG-2025-2026		ECE_5153:Analog VLSI Design	
#	Course Outcome	Description	Marks
1	1	CO 1	10
2	2	CO 2	10
3	3	CO 3	10
4	4	CO 4	10
5	5	CO 5	10
Total			50

* NA - CO Not Selected

Event Name	Subject
058-MIT BLR-MAKEUP EXAM PG-2025-	ECE_5153:Analog VLSI

#	Level	No of Questions	Marks (%)
1	Remember	0	0(0%)
2	Understand	0	0(0%)
3	Application	5	20(40%)
4	Analysis	5	30(60%)
5	Create	0	0(0%)
6	Evaluate	0	0(0%)
	Total	10	50 (100%)